THE REMIX

Synthetic Control & Synthetic DiD

Day 3 of 5: Afternoon

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The problem: ETA terrorism and the Basque economy

In the late 1960s, the separatist group ETA began a campaign of bombings, kidnappings, and assassinations in Basque Country, in northern Spain. By the 1990s, ETA had killed hundreds and was costing the regional economy billions.

Abadie & Gardeazabal (2003) asked: how much did terrorism reduce Basque GDP per capita?

The problem. Spain has 17 autonomous regions. Basque is the only one heavily affected by ETA. You can't run DiD against the average of the other 16 — Basque is different from them on dozens of dimensions *before* terrorism even started.

Abadie, A., and J. Gardeazabal (2003). "The Economic Costs of Conflict: A Case Study of the Basque Country." AER 93(1): 113–132.

The moment the method was born

"I had played with related notions for some time, thinking about panel-data methods to measure the effects of aggregate interventions. But the idea of synthetic controls shaped up in my mind with the Basque example. As always, taking vague concepts to data helps. ***There wasn't much one could do in terms of DD for the Basque example.*** *When I saw what is now Figure 1 in the Basque paper appearing on my computer screen after the code finished running for the first time, the value of the method became immediately clear."*

— Alberto Abadie

The figure he's talking about is on the next slide.

The figure that did it (Basque vs. synthetic Basque)

118 THE AMERICAN ECONOMIC REVIEW MARCH 2003

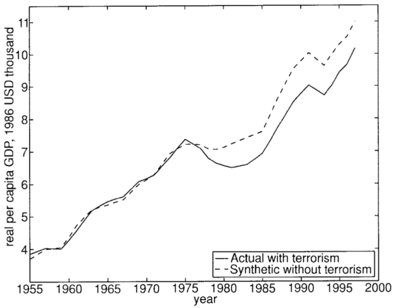


FIGURE 1. PER CAPITA GDP FOR THE BASQUE COUNTRY

Basque GDP per capita (real, dashed) vs. a weighted combination of other Spanish regions chosen to match Basque pre-ETA (solid). The gap is what ETA cost the Basque economy.

This afternoon, two acts

1. **Original synthetic control** — Abadie's method as he wrote it. One treated unit, a donor pool, weights that make a synthetic version of the treated unit from the donors. California Proposition 99 (2010 JASA) as our running example.

2. **Synthetic Difference-in-Differences** — the modern combination of synthetic control with DiD (Arkhangelsky et al. 2021). Weights on units *and* weights on time periods. Demonstrated on the baker.do panel from this morning.

LESSON

1. Original synthetic control

A principled way to build the counterfactual from a donor pool

The core idea: build the counterfactual, don't pick it

Old-school comparative case study. Pick a comparison unit (or a few) that "looks like" the treated unit, and run a DiD.

- Card (1990) used Atlanta, Houston, LA, Tampa as the control for Miami in the Mariel boatlift study.
- Selection is *ad hoc* — the researcher's hand is visible.
- Standard errors don't reflect uncertainty about whether the control reproduces the counterfactual.

Abadie's move. Instead of picking, **construct**. Find a *weighted combination* of every potential donor unit that matches the treated unit on observables before treatment. That convex combination is the "synthetic" counterfactual.

Notation

- $J+1$ units, observed in periods $t = 1, \dots, T$.
- Unit 1 is treated starting at period $T_0 + 1$.
- Units $2, \dots, J+1$ are the **donor pool** — never treated.
- Outcome Y_{it}^0 is what we'd observe *without* treatment. Y_{it}^1 with treatment.

Target parameter. The treatment effect for the treated unit in each post-period:

$$\delta_{1t} \;=\; Y_{1t}^1 - Y_{1t}^0 \;=\; Y_{1t} - Y_{1t}^0 \qquad \text{for } t > T_0.$$

The missing piece is Y_{1t}^0. We'll build it from the donor pool.

The synthetic control: a convex combination of donors

Let $W = (w_2, \ldots, w_{J+1})'$ with $w_j \geq 0$ and $\sum_j w_j = 1$.

The synthetic control's value at time t:

$$\widehat{Y}^0_{1t} = \sum_{j=2}^{J+1} w_j \, Y_{jt}.$$

- Each w_j tells us how much donor unit j contributes to the synthetic version of unit 1.
- Non-negativity rules out extrapolation: the synthetic is in the **convex hull** of the donors.
- Sum-to-one keeps it on the same scale.
- Most weights will be zero in practice — sparse weights $\Rightarrow$ transparent attribution.

Choosing the weights: match the treated unit's pre-period covariates

Let X_1 be a $k \times 1$ vector of pre-treatment characteristics for the treated unit (covariates plus possibly pre-period outcomes). Let X_0 be $k \times J$ for the donors.

The weight problem

$$W^* \;=\; \arg\min_{W} \|X_1 - X_0 W\|_V \;=\; \arg\min_{W} \sqrt{(X_1 - X_0 W)' V (X_1 - X_0 W)}$$

- V is a $k \times k$ diagonal weighting matrix: v_m is the importance of covariate m in the match.
- Constraints: $w_j \geq 0$, $\sum w_j = 1$.
- Quadratic program with linear constraints — a closed-form numerical solver (`synth` in Stata).

The intuition: start with a much simpler version of the problem

Before the convex hull and the constraints, forget weights for a moment.

Suppose you had a single treated unit, and you got to pick *one* control from a pool. Which one would you pick?

The one whose pre-period covariates are closest to the treated unit's. Nearest neighbor. The same idea as synth — just with a discrete choice instead of a weighted blend.

Next: a toy example with 10 trainees and 20 candidate controls.

Worksheet: Google Sheet

A toy example: 10 trainees, 20 candidate controls

Setup.

- 10 trainees (treated): known age and GPA.
- 20 candidates (controls): known age and GPA.
- Procedure: for each trainee, pick *one* control. Compute the gap.

Distance measure. The Euclidean distance between the trainee sample and the picked controls:

$$\text{discrepancy} = \sqrt{\sum_{i=1}^{10} \left[(\text{age}_i - \text{age}_{j(i)})^2 + (\text{gpa}_i - \text{gpa}_{j(i)})^2 \right]}$$

The smaller this number, the better the match.

Pick one set of controls at random: discrepancy $= 45.76$

Random pick #1

45.76

units of pre-period discrepancy

What this number means. The trainee sample and the chosen controls differ by 45.76 “units” in joint (age, GPA) space. *Not* 45.76 years of age and *not* 45.76 GPA points — just 45.76 in the rescaled Euclidean metric.

Can we do better with a different draw? Let’s try.

Pick a different set: discrepancy $= 32.53$

Random pick #1	Random pick #2
45.76	32.53

The second random pick is closer. Two random picks, two different gaps. **The discrepancy depends on *which controls you choose*.**

If random sampling gives different gaps, there must be a best pick. Can we find it?

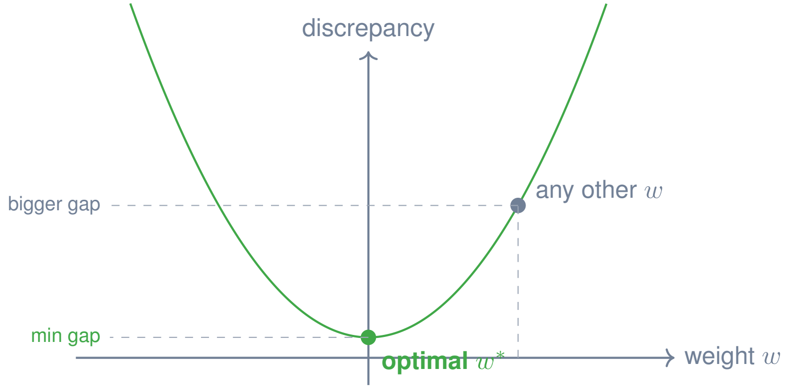
The optimal nearest-neighbor pick: discrepancy $\approx$ 2.99

Pick #1	Pick #2	Optimal
45.76	32.53	**2.99**

Searching over picks dramatically reduces the gap. The optimum is a thousand-fold improvement over random.

This is the minimization. The next slide shows why it works.

Why optimization works: the gap is U-shaped in the weight



The discrepancy is convex in the weight. One minimum. Any other choice is strictly worse.

From discrete picks to continuous weights

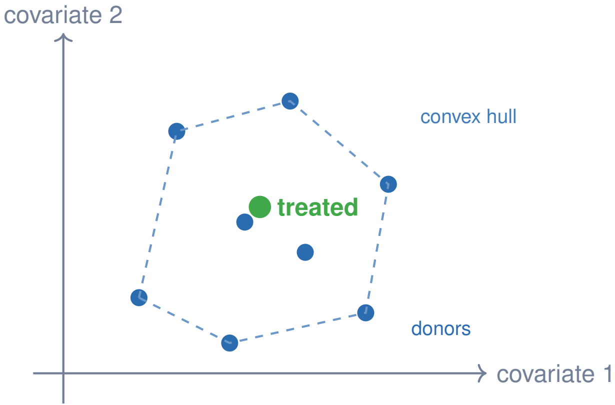
Nearest neighbor. You picked one control per trainee. Discrete choice. The best you could do was 2.99.

Synthetic control. Don't pick one. Build a *weighted blend* of all 20 candidates: w_1 on candidate 1, w_2 on candidate 2, . . ., w_{20} on candidate 20. Weights sum to 1; none negative.

Same minimization problem. The discrepancy is a function of the weight vector W, and we find the W that brings it as close to zero as possible.

Continuous weights give you a strictly richer set of possible "matches" than discrete picks — and the optimum is often much closer to zero.

The convex hull: what the weights can — and cannot — do



Weights $w_j \geq 0$ with $\sum w_j = 1$ produce a point inside the polygon. If the treated unit is inside, the gap can in principle go to zero. If outside, the best match sits on the boundary.

Choosing V: which covariates matter most?

V is the meta-weighting matrix that decides which covariates are most important for the pre-period match.

Three options:

1. **Regression.** Use the variances from a regression of outcome on covariates.
2. **Calibration.** Iterate V until weight choices look right.
3. **Cross-validation (Abadie's recommendation).** Split pre-period into a training and a validation window. For any V, compute $W^*(V)$ on training data, then minimize MSPE on the validation data.

$$\text{MSPE}(V) = \sum_{t=1}^{T_0^{\text{val}}} \left(Y_{1t} - \sum_{j=2}^{J+1} w_j^*(V)\, Y_{jt} \right)^2$$

Cross-validation is what `synth, nested allopt` does by default in Stata.

LESSON

2. California Prop 99

Synthetic control's most famous application

California Proposition 99 (1988)

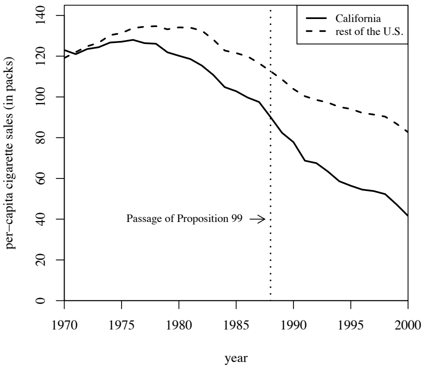
The policy. California passed the first comprehensive state-level tobacco control law in 1988:

- 25-cent-per-pack tax increase
- earmarked revenues for health and anti-smoking budgets
- state-funded media campaigns
- enabled clean-air ordinances throughout the state
- over $100M/year in anti-tobacco spending

The question. Did Prop 99 reduce cigarette consumption in California?

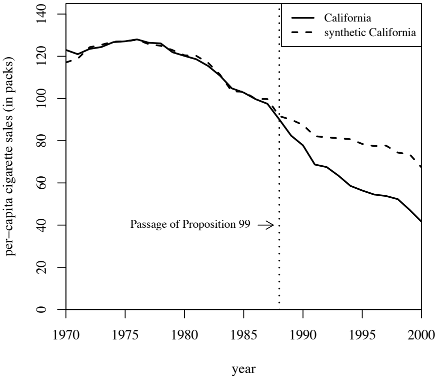
The donor pool. 38 states that didn't pass comparable tobacco controls. Excludes 11 states that adopted similar laws after 1988. *Abadie, A., A. Diamond, J. Hainmueller (2010). JASA 105(490): 493–505.*

California vs. “the rest of the U.S.”



Per-capita cigarette sales. CA is below the rest of the U.S. even before Prop 99 — so a naive “CA minus rest of U.S.” comparison is contaminated by baseline differences. The picture screams for a better counterfactual.

California vs. synthetic California



The weighted combination of donor states (synthetic CA, dashed) tracks CA closely from 1970 to 1988. Post-1988, CA falls ∼ 20+ packs per capita below synthetic CA. ***That gap is the estimated treatment effect.***

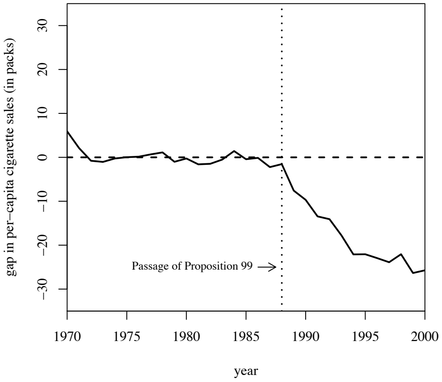
Predictor balance: actual vs. synthetic California

Variables	California Real	California Synthetic	Average of 38 control states
Ln(GDP per capita)	10.08	9.86	9.86
Percent aged 15-24	17.40	17.40	17.29
Retail price	89.42	89.41	87.27
Beer consumption per capita	24.28	24.20	23.75
Cigarette sales per capita 1988	90.10	91.62	114.20
Cigarette sales per capita 1980	120.20	120.43	136.58
Cigarette sales per capita 1975	127.10	126.99	132.81

Note: All variables except lagged cigarette sales are averaged for the 1980-1988 period (beer consumption is averaged 1984-1988).

Synthetic CA matches CA on every predictor in the model — log GDP per capita, retail price of cigarettes, beer consumption, age, and lagged cigarette sales. Much closer than the rest-of-U.S. comparison.

The treatment effect: gap between CA and synthetic CA



The gap series. Flat at zero before 1988 (good match). Grows steadily afterward, reaching ≈ -25 *packs per capita per year by 2000. Prop 99 reduced cigarette consumption substantially.*

Lesson

3. Inference

Placebo permutation tests when you have one treated unit

Inference is awkward with one treated unit

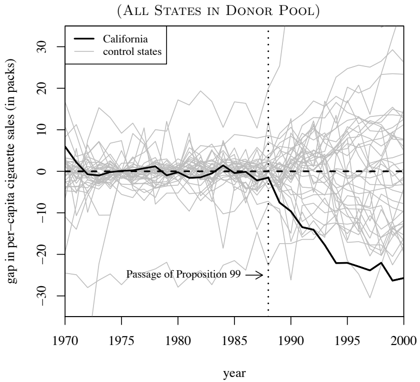
The problem. Classical asymptotics depend on letting the number of treated units grow. Synthetic control was designed for $N_1 = 1$ — there's only *one* California. No clustering, no large-N argument.

The fix (Fisher's sharp null logic).

- Reapply the synthetic-control method to every donor state, pretending *each* donor was treated in 1988.
- Compute the "placebo" gap for each donor.
- Compare California's gap to the distribution of placebo gaps.
- If California's gap is extreme relative to the placebo distribution, the effect is unlikely under no treatment.

*This is **not** a standard p-value — it's an exact randomization-style test under the sharp null.*

California vs. 38 placebo states



California's post-1988 gap (bold) is more extreme than every placebo state. The exact p-value: $1/(J+1) \approx 0.026$. The picture is the inference.

Refinement: the post-to-pre RMSPE ratio

The concern. Some donor states have a *bad* pre-period fit — the placebo trick attributes big post-period gaps to them just because their pre-period was noisy.

The test statistic.

$$\text{RMSPE post}/\text{RMSPE pre} = \frac{\sqrt{\frac{1}{T-T_0}\sum_{t>T_0}\left(Y_{1t}-\sum_j w_j^* Y_{jt}\right)^2}}{\sqrt{\frac{1}{T_0}\sum_{t\leq T_0}\left(Y_{1t}-\sum_j w_j^* Y_{jt}\right)^2}}$$

- Numerator: how big is the post-period gap?
- Denominator: how big was the pre-period gap?
- Big ratio $\Rightarrow$ a real treatment effect, not just noise.

The p-value is California's RMSPE-ratio rank divided by $J+1$.

Lesson

4. When SC works (and when it doesn't)

The conditions, and the bridge to synthetic DiD

What SC actually assumes: a factor model

The untreated potential outcome follows:

$$Y_{jt}(0) \;=\; \alpha_j \;+\; \tau_t \;+\; \beta_t X_j \;+\; \gamma_t Z_j \;+\; \varepsilon_{jt}$$

- α_j — unit fixed effect; τ_t — common time trend.
- X_j — *observed* covariates; β_t time-varying loading.
- Z_j — *unobserved* heterogeneity; γ_t time-varying return.
- ε_{jt} — idiosyncratic noise.

TWFE DiD requires $\gamma_t = \gamma$ for all t — PT in every period, pre *and* post. SC lets γ_t move.

Why a long pre-period: structure vs. noise

Two terms in $Y_{jt}(0)$ vary across j and t:

- $\gamma_t Z_j$ — **structure.** Persistent. The same Z_j shows up year after year, just with a different return γ_t.
- ε_{jt} — **noise.** Independent across periods.

SC fits weights on the *pre-period* data. We want those weights to track $\gamma_t Z_j$, not the noise.

Short T_0: the noise looks like signal. The fit overfits ε.

Long T_0: noise averages out across periods. The fit picks up the persistent Z_j loading.

Long pre-period $\Rightarrow$ match the structure, not the noise.

Abadie's bias bound (2021 JEL, Sec. 3.3) makes this precise: bias is decreasing in T_0, increasing in Var(ε) and in the number of unobserved factors in Z_j.

Conditions for SC to do its job

1. **Convex-hull condition.** The treated unit's pre-period covariates lie within the convex hull of the donors. If California is extreme on every dimension, no weighted combination of donors can match it.
2. **Long pre-period.** Enough T_0 to separate $\gamma_t Z_j$ from ε_{jt}. Short pre-period $\Rightarrow$ overfit the noise.
3. **Similar donor pool.** Donors should look like the treated unit on X_j *and* plausibly on Z_j. A bigger pool of dissimilar units inflates the bias bound, not shrinks it.
4. **Good pre-period fit.** Pre-period RMSPE small relative to the post-period gap. If the synthetic doesn't track the unit *before* treatment, nothing after is credible.

What if the conditions fail?

Three real failure modes:

- **The treated unit is extreme.** California's pre-period values may sit at the edge of the donor distribution. The convex hull doesn't reach it. The synthetic can only approximate.
- **The pre-period is short.** Not enough periods to average out ε. The weights chase noise.
- **Many treated units.** The original method is built for $N_1 = 1$. If you have 10 treated states, do you run 10 separate fits? Combine them how?

Synthetic DiD relaxes (1) via a DiD-style level shift and gains efficiency from time weights.

For (3): penalized synth (Abadie–L'Hour 2021), microsynth (Robbins et al. 2017), augmented SC (Ben-Michael et al. 2021).

Lesson

Act II—Synthetic Difference-in-Differences

Arkhangelsky, Athey, Hirshberg, Imbens, Wager (2021)

LESSON

5. Why combine SC and DiD?

Their weaknesses are complementary

DiD and SC each have a problem the other can fix

DiD assumes parallel trends.

- If treated and controls are on different trajectories pre-treatment, DiD is biased.
- Pre-trend tests don't really test PT (Roth 2022, from this morning).

Synthetic control matches pre-trends.

- Picks weights to match pre-period covariates.
- But: only one treated unit; needs the convex-hull condition; struggles with short pre-period.

What if you used *weights* like SC *and differencing* like DiD? That's the idea behind synthetic DiD.

Arkhangelsky, D., S. Athey, D. Hirshberg, G. Imbens, S. Wager (2021). "Synthetic Difference-in-Differences." AER 111(12): 4088–4118.

The big idea: weights on units *and* time

SC gives you unit weights w_i that build a synthetic version of the treated unit.

DiD subtracts a time effect β_t that's common across units.

SDID adds time weights λ_t. Pre-periods that are most predictive of the post-period get higher weight; less informative pre-periods get less.

The combination

A two-way fixed-effects regression **weighted by both w_i and λ_t.**

The original SC has unit weights and no time weights (or, more precisely, an implicit equal-weighting). SDID makes the time weights an estimation target.

Three estimators: SC, DiD, SDID

SC: unit weights, no unit FE, no time weights.

$$\widehat{\tau}^{\text{SC}} = \arg\min_{\tau,\lambda_t} \sum_{i,t} (Y_{it} - \lambda_t - \tau D_{it})^2 \, w_i^{\text{SC}}$$

DiD: unit FE, time FE, equal weights.

$$\widehat{\tau}^{\text{DiD}} = \arg\min_{\tau,\mu,\alpha_i,\beta_t} \sum_{i,t} (Y_{it} - \mu - \alpha_i - \beta_t - \tau D_{it})^2$$

SDID: unit FE, time FE, *plus* unit weights w_i and time weights λ_t.

$$\widehat{\tau}^{\text{SDID}} = \arg\min_{\tau,\mu,\alpha_i,\beta_t} \sum_{i,t} (Y_{it} - \mu - \alpha_i - \beta_t - \tau D_{it})^2 \, w_i \, \lambda_t$$

SDID is DiD with both kinds of weights bolted on.

SDID nests both SC and DiD

Set parts of SDID to defaults and you recover the other estimators exactly.

- **Additive systematic component** ($L_{it} = \alpha_i + \beta_t$): oracle $\widetilde{w}_i \to 1/N_{co}$ and $\widetilde{\lambda}_t \to 1/T_{pre}$. **SDID $\to$ DiD asymptotically.**
- **Set $w_0 = 0$ and $\zeta = 0$** (no intercept, no ridge) with $N_{tr} = 1$: the Abadie–Diamond–Hainmueller weight problem exactly. **SDID $\to$ SC.**

Reading. SC, DiD, and SDID are one estimator at three settings of (w_0, ζ, λ). Bad pre-trends? Time weights kick in. Sharp pre-trend match? Unit weights collapse to SC. Additive L? Weights go uniform and you have DiD.

Prop 99 revisited: what each estimator says

Same California data we just used in §2 — **five estimators, five answers** (paper Table 1, p. 8).

Method	**Estimate** (packs/year)	**SE**
SDID	-15.6	(8.4)
SC (Abadie et al. 2010)	-19.6	(9.9)
DID	-27.3	(17.7)
MC (matrix completion)	-20.2	(11.5)
DIFP (Doudchenko–Imbens)	-11.1	(9.5)

- SDID sits *between* SC and DIFP — SC's tight pre-period match *and* a DiD-style intercept.
- **SDID's SE (8.4) is the smallest** despite being the more flexible estimator. Locality of weighting strips out predictable variation.

LESSON

6. Estimating the weights

Where do w_i and λ_t come from?

Unit weights $\widehat{w}_i$: match the pre-period

Same idea as Abadie's SC, with a regularization term:

$$\widehat{w} \;=\; \arg\min_{w_0, w} \sum_{t=1}^{T_0} \left(w_0 + \textstyle\sum_{i \in \text{control}} w_i \, Y_{it} - \frac{1}{N_1} \sum_{i \in \text{treated}} Y_{it}\right)^2 \;+\; \zeta^2 T_0 \, \|w\|_2^2$$

- Find weights so the weighted-control pre-period average matches the treated pre-period average.
- Constraints: $w_i \geq 0$, $\sum w_i = 1$.
- Allows an intercept w_0 (SC doesn't): SDID matches pre-period *trends*, not levels.
- $\zeta^2 T_0 \, \|w\|_2^2$ is a ridge penalty tied to the variance of one-period changes in untreated units. Prevents overfit; spreads weight across donors.

Time weights $\widehat{\lambda}_t$: which pre-periods are predictive?

Analogous, but in the time dimension:

$$\widehat{\lambda} = \arg\min_{\lambda_0,\lambda} \sum_{i\in\text{control}} \left(\lambda_0 + \textstyle\sum_{t\le T_0} \lambda_t Y_{it} - \frac{1}{T-T_0} \sum_{t>T_0} Y_{it}\right)^2$$

- Pre-periods get weight in proportion to how well they predict each control unit's *post*-period average.
- Constraints: $\lambda_t \ge 0$, $\sum \lambda_t = 1$.
- No substantive regularization on time weights. *Asymmetry is structural: errors are assumed independent across units but serially correlated within unit — there's no within-time cross-unit correlation to ridge away.*

This is the SDID-specific move. Plain DiD treats all pre-periods equally; SDID prioritizes the ones that carry predictive information.

The SDID procedure, in three steps

1. **Estimate unit weights** $\widehat{w}_i$ via the pre-period matching problem.
2. **Estimate time weights** $\widehat{\lambda}_t$ via the per-control predictive problem.
3. **Run weighted two-way fixed effects** using $\widehat{w}_i \cdot \widehat{\lambda}_t$ as the cell weight, with treatment indicator D_{it}.

The coefficient on D_{it} is the SDID ATT.

Software: `sdid` (Stata, SSC); `synthdid` (R; github.com/synth-inference/synthdid). Standard errors via three options: jackknife, bootstrap, placebo.

LESSON

7. Why does SDID work?

The doubly-robust property

The bias decomposition: where does SDID get its robustness?

For the SDID estimator and the “oracle” weights $(\widetilde{w}, \widetilde{\lambda})$ that perfectly balance the systematic component:

$$\widehat{\tau}^{\text{SDID}} - \tau = \underbrace{\varepsilon(\widetilde{w}, \widetilde{\lambda})}_{\text{oracle noise}} + \underbrace{B(\widetilde{w}, \widetilde{\lambda})}_{\text{oracle bias}} + \underbrace{\widehat{\tau}(\widehat{w}, \widehat{\lambda}) - \widehat{\tau}(\widetilde{w}, \widetilde{\lambda})}_{\text{deviation from oracle}}$$

- Oracle noise: shrinks with sample size.
- Oracle bias: zero if either unit or time weights balance the systematic component.
- Deviation: shrinks as estimated weights approach oracle weights under regularization.

If *either* the unit weights or the time weights work, SDID is consistent. That’s doubly robust.

Same idea as yesterday's DR-DiD — just with a second axis

Yesterday afternoon (DR-DiD, Sant'Anna-Zhao 2020):

- Outcome regression model on X + propensity score weighting on X.
- Consistent if *either* model is correct.
- One axis of robustness: the covariate X.

SDID:

- Unit weights match pre-period *across units* + time weights balance pre-period *across time*.
- Consistent if *either* set of weights captures the systematic component.
- Two axes of robustness: unit and time.

Two-dimensional doubly robust. Yesterday afternoon's lesson, applied to the panel structure.

LESSON

8. SDID on baker.do

Demonstration: does SDID recover the truth?

Pre-flight check: SDID needs *block* treatment

The paper's main result (Theorem 1) assumes block assignment: $W_{it} = \mathbf{1}\{i > N_{co},\, t > T_{pre}\}$. All treated units start treatment at the same time. No staggered adoption in the headline.

So what about staggered settings (like baker.do)?

Appendix VI (pp. 41–42): split a staggered panel into block sub-panels (one per adoption date), run SDID on each, take a weighted average (weights $\propto$ fraction of treated unit-period pairs).

Practical implication for baker.do: drop the always-treated 2004 cohort to create a clean block (1986/1992/1998 treated vs. never-treated). Same restriction we used for CS this morning. For richer multi-cohort designs, Ben-Michael, Feller & Rothstein (2019) is the cousin paper.

Recall this morning's simulation (baker.do)

- 1,000 firms, 4 cohorts of states, treated at 1986/1992/1998/2004.
- Dynamic treatment effects: ATT grows each year since treatment.
- Simple average across post-period treated cells: **truth** $= +82$.
- TWFE returned -6.69. CS (DR-DiD) on the feasible sample returned $+68.34$.

Question. What does SDID give on the same data?

We're running the headline regression both ways — once on the universe-treated panel, once on the pre-2004 feasible sample where a never-treated comparison still exists.

The SDID code, in five lines

```
* Install once, then load baker.dta
ssc install sdid, replace

* Drop the always-treated 2004 cohort so we have a feasible never-treated group
drop if treat_date == 2004

* Run SDID with the treatment indicator
sdid y state year treat, vce(bootstrap) reps(200) graph
```

What the command does:

- Estimates $\widehat{\omega}_i$ to match pre-period across units.
- Estimates $\widehat{\lambda}_t$ to weight informative pre-periods.
- Runs the weighted TWFE regression and reports $\widehat{\tau}^{\text{SDID}}$ with bootstrap SE.
- `graph` produces the unit-and-time-weighted parallel-trends picture.

Software: `sdid` (Pailañir & Clarke, Stata SSC). R: `synthdid`.

The headline: SDID recovers the truth too

	Truth (feasible sample)	Estimate
TWFE	68.33	26.81^{***}
CS (DR-DiD)	68.33	68.34^{***}
SDID	68.33	$\mathbf{67.9}^{***}$

- TWFE still attenuates: the variance weights and forbidden 2$\times$2s carry over.
- CS and SDID both land essentially on the truth.
- SDID gets there a different way: *weighting the panel*, not modeling each ATT(g, t) separately.

Indicative numbers from baker.do dynamic-effects DGP under feasible-sample restriction. Exact value depends on bootstrap seed.

LESSON

9. Practical considerations

Where SDID earns its keep, and where you should pause

When to reach for SDID

- **A *block* of treated units.** SC was designed for $N_1 = 1$ (and runs unit-by-unit beyond that). SDID is designed for a block of treated states treated jointly at the same time.
- **Imperfect pre-period match.** If SC's convex hull doesn't include the treated unit, SDID's w_0 intercept can soak up the level gap and match trends.
- **Suspect parallel trends.** Time weights downweight pre-periods that don't predict the post-period — automatic robustness against pre-trends.
- **Long panel with many states.** The setting where SDID's regularization kicks in.

Where to be careful with SDID

- **Check weight concentration.** If a few donors get nearly all the unit weight, you've fit a clever 2×2. Report the weights.
- **Visually inspect pre-trends after weighting.** The pre-period match *after* applying $\widehat{\omega}_i$ and $\widehat{\lambda}_t$ is what matters. The `sdid, graph` option produces it.
- **Standard errors — depends on N_{tr}.** With $N_{tr} = 1$ (e.g., Prop 99) only **placebo** is defined (jackknife collapses). Small $N_{tr} > 1$: bootstrap is unstable; jackknife (conservative) and placebo are alternatives. Large N_{tr}: bootstrap is preferred.
- **Heterogeneous effects.** SDID reports an ATT. If you suspect substantial heterogeneity, also run CS by group.

LESSON

10. Synthesis

SC, DiD, SDID — when to use which

Which estimator should I use?

The choice between SC, DiD, and SDID follows from three things you can see *before* running anything.

1. **Treatment timing.** One treated unit? A block? Staggered cohorts?
2. **Pre-trends.** Parallel in raw data? Different in level? Different in slope?
3. **Specific concerns.** Heterogeneous effects? Dynamic effects? Inference with $N_{tr} = 1$?

Three lenses — three slides — one decision.

The lenses interact. Read all three before choosing. The decision matrix at the end consolidates.

Lens 1: how many treated units, and when?

Count the treated units and look at adoption dates. The shape of W_{it} drives the choice.

Setting	First-choice estimator
$N_{tr} = 1$, long pre-period	**SC** (or SDID with placebo SE)
$N_{tr} > 1$, single adoption date (block)	**SDID**
$N_{tr} > 1$, staggered adoption	**CS** / **dCdH** (or SDID block-by-block, App. VI)
N_{tr} very large, parallel trends credible	**DiD**

- SDID's headline theorem is for *block* assignment. Staggered settings: the Bacon-decomp issues from this morning still apply.
- DiD is right when pre-trends genuinely line up — don't over-engineer.

Lens 2: what do the pre-trends look like?

Plot the raw pre-trend lines. What you see tells you which assumption is cheap and which is binding.

Pre-period pattern	Use
Parallel slopes, similar levels	**DiD** (cheapest assumption)
Parallel slopes, different levels	**SDID** (w_0 absorbs the gap)
Non-parallel; donor reweighting matches exactly	**SC** (if convex hull holds)
Non-parallel; donor pool doesn't bracket treated	**SDID** or augmented SC
Pre-period too short to read trend	No estimator saves a bad design

If pre-trends visibly diverge, DiD is the wrong tool. The question is which weighting fix.

Lens 3: what's your specific concern?

Match the worry to the estimator that addresses it.

Specific concern	Use
Heterogeneous effects across cohorts	**CS** (reports ATT(g, t))
Dynamic effects growing over time	CS event study; or **synthdid** ES aggregation
$N_{tr} = 1$, need inference	**SC** or **SDID with placebo**
Time-varying confounders X_{it}	**DR-DiD** (yesterday); or residualize first
Audience needs transparent weights	**SC** (sparse donor pool visible)
Long panel, many states, suspect PT	**SDID**

Lenses interact. $N_{tr} = 1$ *(Lens 1) plus parallel trends in raw data (Lens 2) means SC and DiD both work — Lens 3 (transparency? inference? heterogeneity?) breaks the tie.*

Three estimators, one decision matrix

	Synthetic Control	**DiD / CS**	**SDID**
Treated units	One (or per-unit)	Many	Many, block-treated
Pre-trends	Match by weights	Assume parallel	Weighted-match
Time weights	Equal	Equal	Optimized
Inference	Placebo permutation	Bootstrap / SE	Bootstrap / jackknife / placebo
Strength	Transparent counterfactual	Flexible aggregation	Robust to weak PT
Weakness	Convex-hull binding	PT untestable	Black-boxier weights

Decision rule:

- One treated unit, long pre-period, donors that bracket it $\Rightarrow$ **SC**.
- Many treated units, parallel trends look credible $\Rightarrow$ **CS** / **DR-DiD**.
- Many treated units, parallel trends look shaky, long pre-period $\Rightarrow$ **SDID**.

What ties this afternoon to the rest of the week

- **Day 1.** The 2$\times$2 DiD — forward-engineer from ATT and PT.
- **Day 2 morning.** Triple-diff when PT fails.
- **Day 2 afternoon.** Covariates: OR, IPW, doubly-robust.
- **Day 3 morning.** Staggered: Bacon, CS, SA, dCdH, BJS, Honest DiD.
- **Day 3 afternoon.** Synthetic control: weights, not models. Then SDID: weights on units *and* time.

One sentence: match the counterfactual to the unit (and the time), or trust an assumption you cannot test. The rest is engineering.

Reading for tonight: Arkhangelsky et al. (2021); the practitioner's guide §A.5 for the augmented and matrix-completion extensions.

Lab: rerun the Brazil checklist with SDID

The morning lab (Brazil mental-health reform) ran the 9-step checklist with CS as the headline estimator. Now do it with SDID and compare.

1. Drop the 2002 always-treated cohort (same restriction as morning).
2. Run `sdid sim_agressao cod ano treat` with bootstrap SE.
3. Compare the SDID ATT to the CS ATT from the morning.
4. Inspect `sdid, graph`: do the weighted pre-trends look parallel?
5. Inspect the unit-weight distribution: which donor municipalities carry the synthetic Basque-equivalent?
6. Report which estimator you'd prefer for the Brazil setting and why.

Walkthrough: `glasgow/labs/Brazil/README.md` (extend the existing 9 steps with an SDID step 10). Data still at GitHub `mixtape/brazil.dta`.

Three days, one rule.

Pick the parameter first.

Build the counterfactual second.
Trust an assumption only when you have to.

Scott Cunningham · University of Glasgow · May 20, 2026